



Dissolved Oxygen and Water

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By [Water Science School](#)

June 5, 2018

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Dissolved oxygen (DO) is a measure of how much oxygen is dissolved in the water – the amount of oxygen available to living aquatic organisms. The amount of dissolved oxygen in a stream or lake can tell us a lot about its water quality.

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The USGS has been measuring water for decades. Some measurements, such as [temperature](#), [pH](#), and [specific conductance](#) are taken almost every time water is sampled and investigated, no matter where in the U.S. the water is being studied. Another common measurement often taken is dissolved oxygen (DO), which is a measure of how much oxygen is dissolved in the water – DO can tell us a lot about water quality.

Dissolved Oxygen and Water

Although water molecules contain an oxygen atom, this oxygen is not what is needed by aquatic organisms living in natural waters. A small amount of oxygen, up to about ten molecules of oxygen per million of water, is actually dissolved in water. Oxygen enters a stream mainly from the atmosphere and, in areas where groundwater discharge into streams is a large portion of streamflow, from [groundwater discharge](#). This dissolved oxygen is breathed by fish and zooplankton and is needed by them to survive.

Dissolved oxygen and water quality

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Rapidly moving water, such as in a mountain stream or large river, tends to contain a lot of dissolved oxygen, whereas stagnant water contains less. **Bacteria in water** can consume oxygen as organic matter decays. Thus, excess organic material in lakes and rivers can cause eutrophic conditions, which is an oxygen-deficient situation that can cause a water body to "die." Aquatic life can have a hard time in stagnant water that has a lot of rotting, organic material in it, especially in summer (the concentration of dissolved oxygen is inversely related to **water temperature**), when dissolved-oxygen levels are at a seasonal low. Water near the surface of the lake— the epilimnion— is too warm for them, while water near the bottom—the hypolimnion— has too little oxygen. Conditions may become especially serious during a period of hot, calm weather, resulting in the loss of many fish. You may have heard about summertime fish kills in local lakes that likely result from this problem.

(Source: [A Citizen's Guide to Understanding and Monitoring Lakes and Streams](#))



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USGS scientist is measuring various water-quality conditions in Holes Creek at Huffman Park in Kettering, Ohio.



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A eutrophic lake where dissolved-oxygen concentrations are low. Algal blooms can occur under such conditions.

process affecting the dissolved-oxygen/temperature relation; water clarity and strength and duration of sunlight, in turn, affect the rate of photosynthesis.

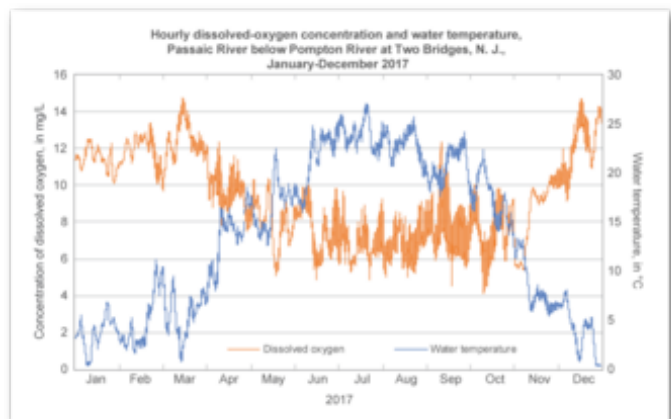
Hypoxia and "Dead zones"

You may have heard about a Gulf of America "dead zone" in areas of the Gulf south of Louisiana, where the Mississippi and Atchafalaya Rivers discharge. A dead zone forms seasonally in the northern Gulf of America when subsurface waters become depleted in dissolved oxygen and cannot support most life. The zone forms west of the Mississippi Delta over the continental shelf off Louisiana and sometimes extends off Texas. The oxygen depletion begins in late spring, increases in summer, and ends in the fall.

Dissolved oxygen, temperature, and aquatic life

As the chart shows, the concentration of dissolved oxygen in surface water is affected by temperature and has both a seasonal and a daily cycle. Cold water can hold more dissolved oxygen than warm water. In winter and early spring, when the water temperature is low, the dissolved oxygen concentration is high. In summer and fall, when the water temperature is high, the dissolved-oxygen concentration is often lower.

Dissolved oxygen in surface water is used by all forms of aquatic life; therefore, this constituent typically is measured to assess the "health" of lakes and streams. Oxygen enters a stream from the **atmosphere** and from **groundwater discharge**. The contribution of oxygen from groundwater discharge is significant, however, only in areas where groundwater is a large component of streamflow, such as in areas of **glacial deposits**. Photosynthesis is the primary



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Water temperature affects dissolved-oxygen concentrations in a river or water body.

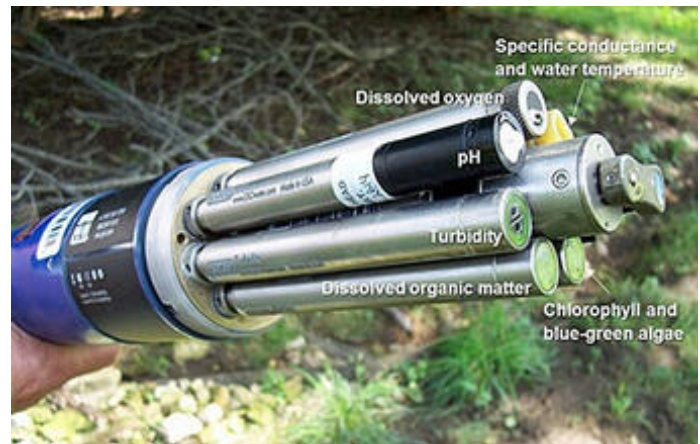
The formation of oxygen-depleted subsurface waters has been associated with nutrient-rich (**nitrogen** and **phosphorus**) discharge from the Mississippi and Atchafalaya Rivers. Bio-available nutrients in the discharge can stimulate algal blooms, which die and are eaten by bacteria, depleting the oxygen in the subsurface water. The oxygen content of surface waters of normal **salinity** in the summer is typically more than 8 milligrams per liter (8 mg/L); when oxygen concentrations are less than 2 mg/L, the water is defined as hypoxic (CENR, 2000). The hypoxia kills many organisms that cannot escape, and thus the hypoxic zone is informally known as the “dead zone.”

The hypoxic zone in the northern Gulf of America is in the center of a productive and valuable fishery. The increased frequency and expansion of hypoxic zones have become an important economic and environmental issue to commercial and recreational users of the fishery.

Measuring dissolved oxygen

Field and lab meters to measure dissolved oxygen have been around for a long time. As this picture shows, modern meters are small and highly electronic. They still use a probe, which is located at the end of the cable. Dissolved oxygen is dependent on temperature (an inverse relation), so the meter must be calibrated properly before each use.

Do you want to test your local **water quality**?



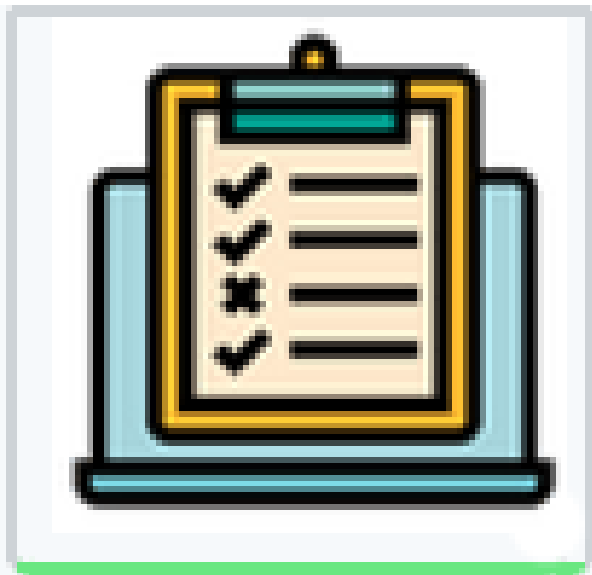
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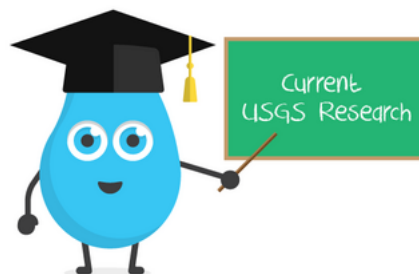
Multi-parameter monitor used to record water-quality measurements.

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